Assignment BIT_ITI420_F12

Deadline: 23:59 O'clock (Damascus Time) on Thursday 4th July 2013

The following problem tackles three main subjects:

- IP addressing.
- VLAN.
- Routing concepts.

Problem:

Let **SCC (SVU Consulting Co)** a company of 1200 employees. This company is composed of four departments:

- **Sales**: contains 200 employees.
- **HR**: contains 200 employees.
- **Technical**: contains 600 employees.
- **Administration**: contains 200 employees.

We suppose that SCC has chosen the class A network id: 10.0.0.0/8 as its main network address. Technical guys decided to divide the main network into four port-based VLANs: 10,20,30 and 40. Fig. 1 represents the connected VLANs when looking at the topology as a Layer 2.

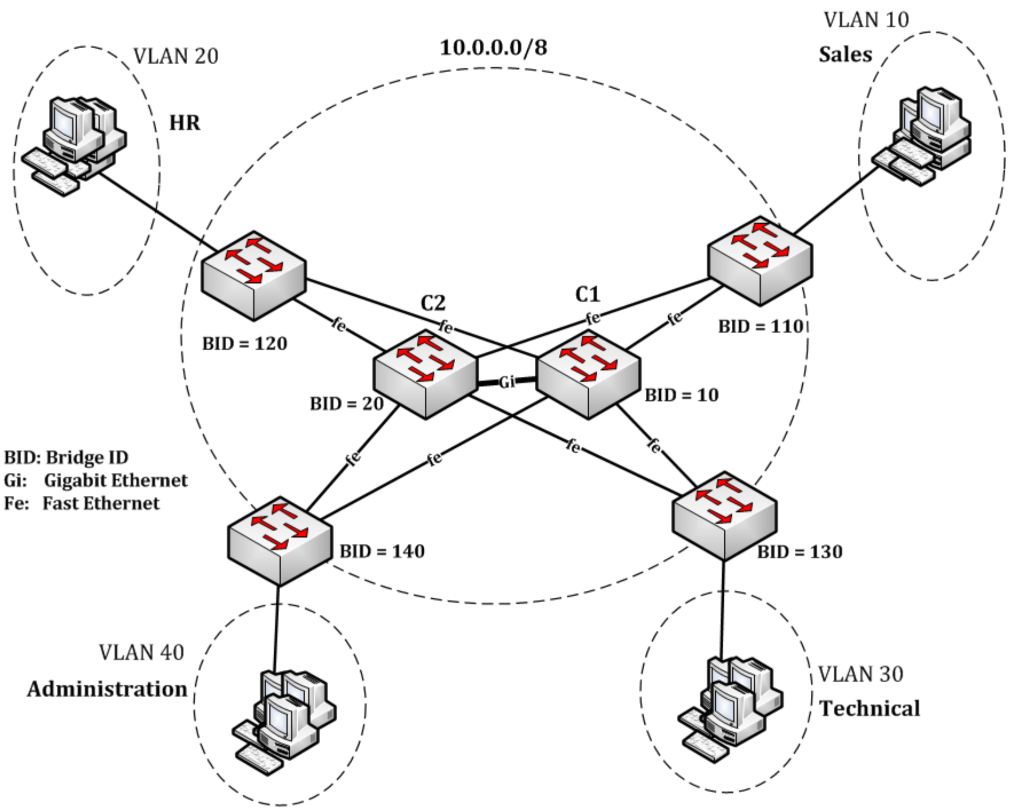

Fig. 1

I. Addressing: 30 Points

1. Suggest an addressing plan that takes into consideration a future expansion of 200% in the number of employees of each department.
2. Give the subnet id of each department.
3. Give the broadcast address of each department.

II. Virtual LANs: 30 Points

Fig.1 shows layer 2 switches isolating the four VLANs.

4. What are the two main advantages of splitting the main network into multiple VLANs?
5. What is the advantage of connecting each departmental switch using two links with the core switches C1 and C2?
6. What are the switch port types of switches C1 and C2? What are the switch port types of the switch “sales”? What is the difference between the two types?
7. What is the main challenge of connecting switches in this way?
8. Describe, briefly, a solution to overcome the above challenge.
9. An IT consultant suggests dividing the network into MAC-based VLANs instead of port-based.
    a. Compare between the two types.
    b. Describe briefly the steps to set up MAC-based VLANs.

III. Routing: 40 Points

Fig. 2 shows the network topology as a Layer 3.

10. Show the routing table of R1, supposing that we assign the last address of each subnet to the Router’s interfaces.
11. Do you need to add any other routes to make the connectivity possible between the four VLANs? Why?

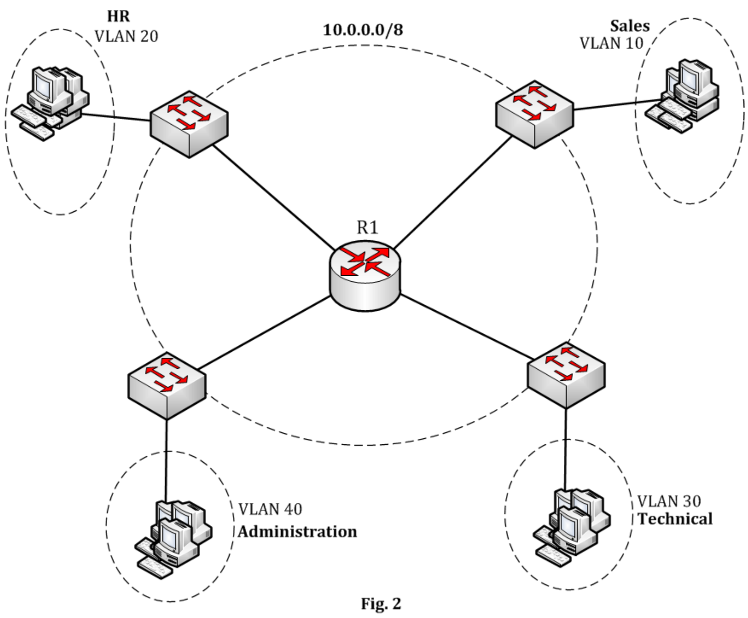

Fig. 2

12. A PC from VLAN10 pings another PC in VLAN 20, and another PC between the switch "sales" and the central router is running a protocol analyzer.
    a. Give the L2 source/destination addresses and L3 source/destination addresses for an outgoing ***ping*** packet.
    b. Represent the protocol encapsulation for the ***ping*** packet as in TCP/IP model. You should determine the header length of each protocol.
13. Let LAN2 a network that is connected via R2 with R1 "central router", as shown in Fig. 3.
    a. Suggest a /30 subnet to the link between R1 and R2.
    b. List the routes to be added on both routers to make the connectivity possible between LAN2 and the four VLANs in SCC Company.

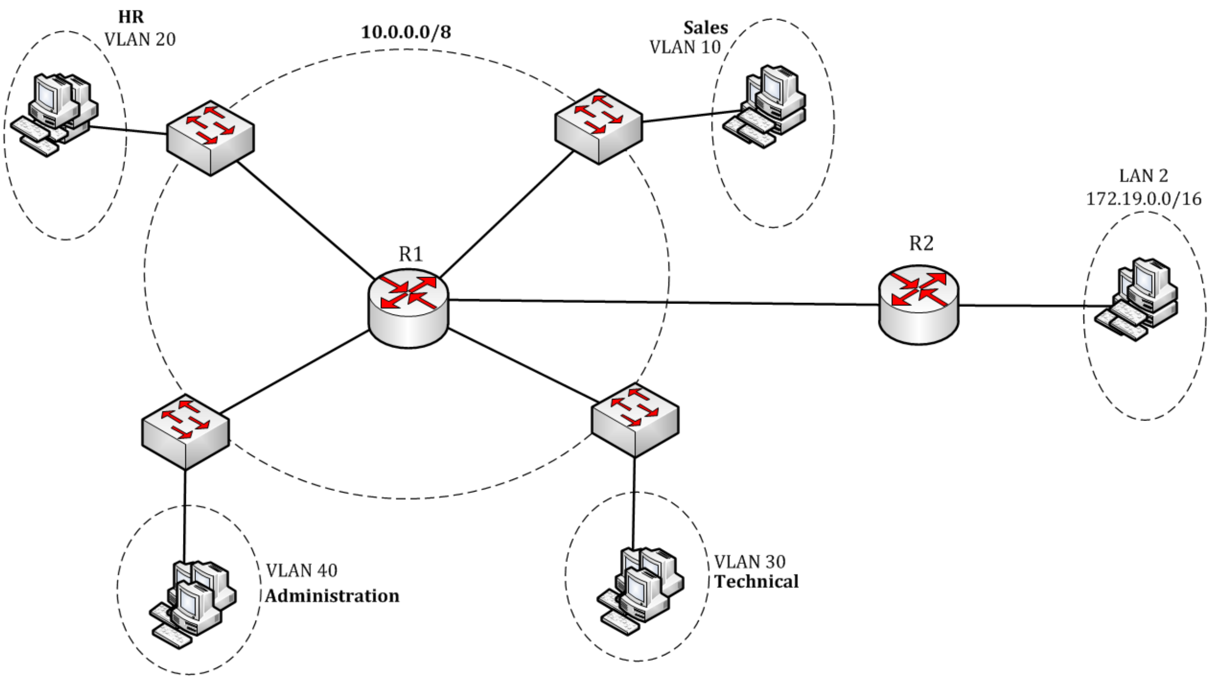

Fig. 3